DIGITAL LOGIC DESIGN · COURSE PROJECT

Smart Parking Slot Availability & Entry Control

A combinational logic architecture with priority encoding

4-Slot Detection · Priority Direction · Automated Gate Release

Presented by

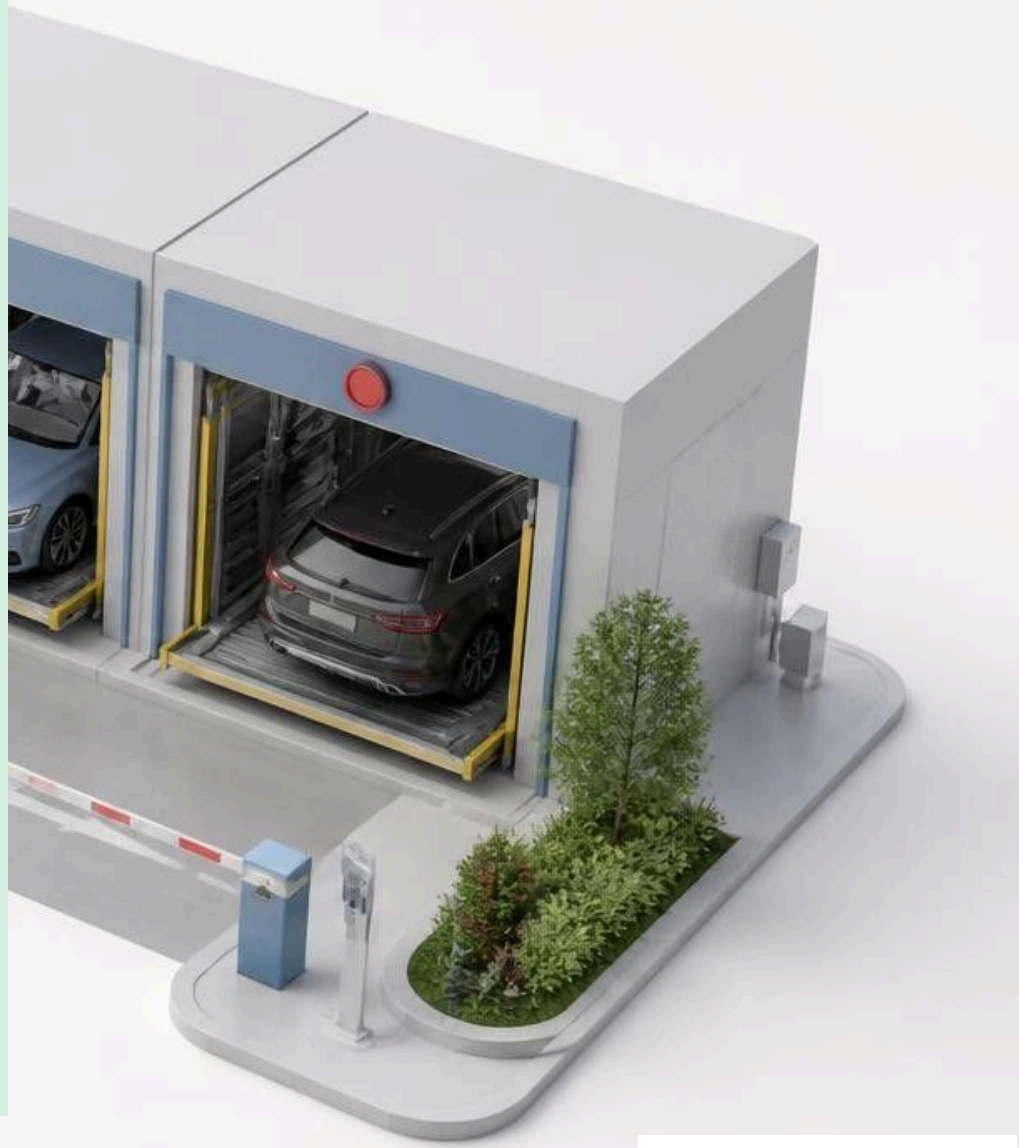
S.Jethendra Vardhan-2620030600

Laxmi Pragya-2620030602

Veda Chandra-2620030606

Chandra Paul-2620030636

Srikrishna-2620030675



LOGIC FOUNDATION

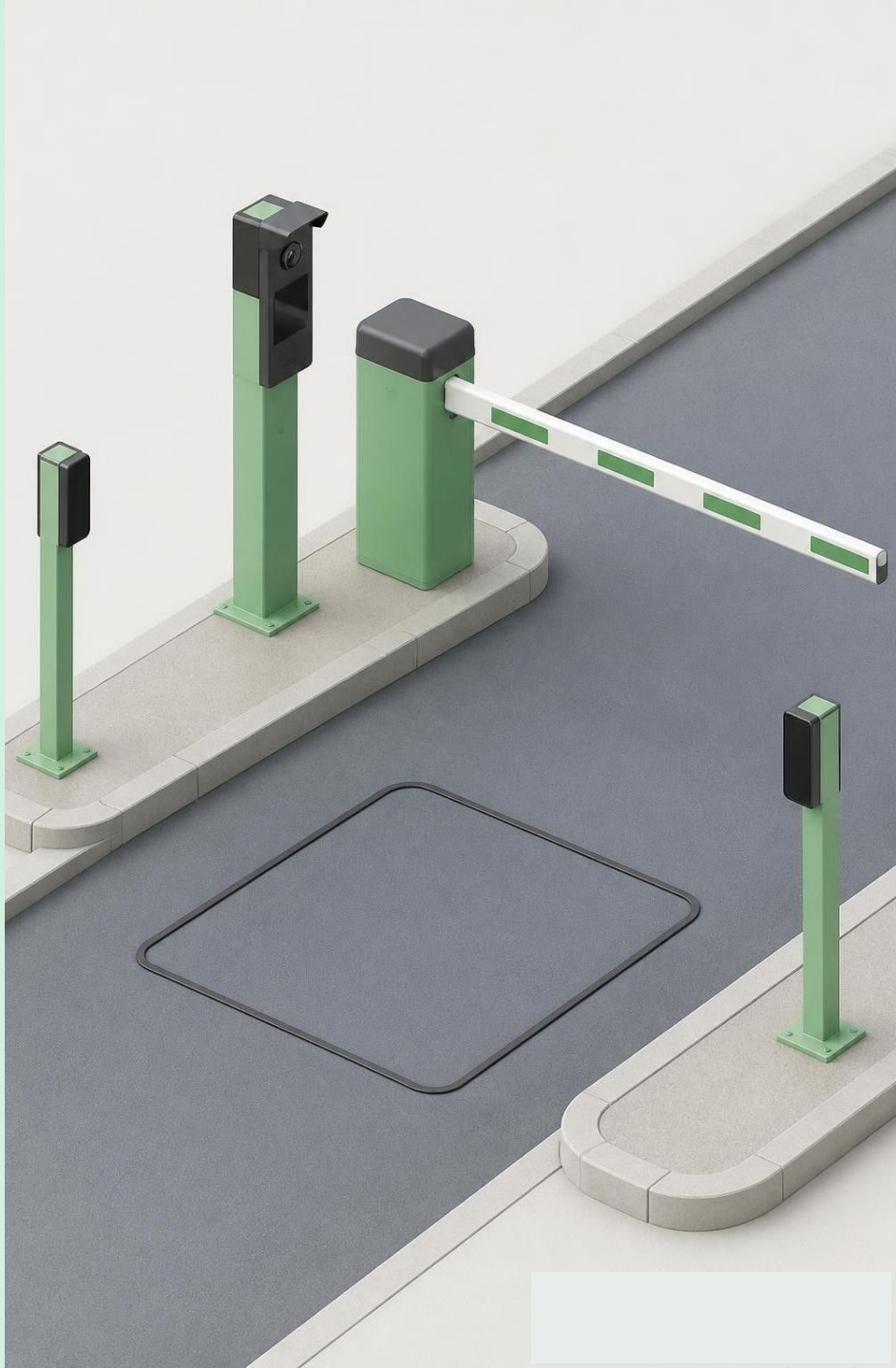
Pure combinational logic needs zero memory

1. **Problem** · Drivers circle occupied lots aimlessly without real-time bay visibility.

2. **Logic premise** · System outputs depend strictly on present inputs; no clock or flip-flops needed.

3. **Project aim** · Design, minimize, and simulate a 4-slot availability detector and gate controller.

RESPONSE TIME DEFINED SOLELY BY GATE PROPAGATION DELAYS



MILESTONES 01 - 03

Objectives and learning outcomes

PHASE 01

Formal mapping

Translate real physical requirements into exact Boolean variables, truth tables, and boundary conditions.

PHASE 02

Gate minimization

Apply Karnaugh maps and algebraic reductions to minimize gate count, IC footprints, and propagation delays.

PHASE 03

Logisim verification

Construct schematic circuits and validate every corner case systematically against theoretical truth tables.

Four sensors, one gate request, three controls

I/O SPECIFICATION

Inputs

S0 to S3 slot sensors (1 = occupied)

V vehicle at gate · P valid permit

Outputs

Slot-free flags F0 to F3 · FULL indicator

GATE open · 2-bit nearest-slot code

BLOCK DIAGRAM

Sensors

S0 to S3, V and P

Availability Logic

Slot-free flags and FULL

Priority Encoder

Nearest-slot code

Entry Controller

GATE open

Display

Slot status and code

SUBSYSTEM 01

Slot availability: free is NOT occupied

Each bay sensor outputs $S_i = 1$ when a vehicle is detected. Inverting gives immediate slot vacancy:

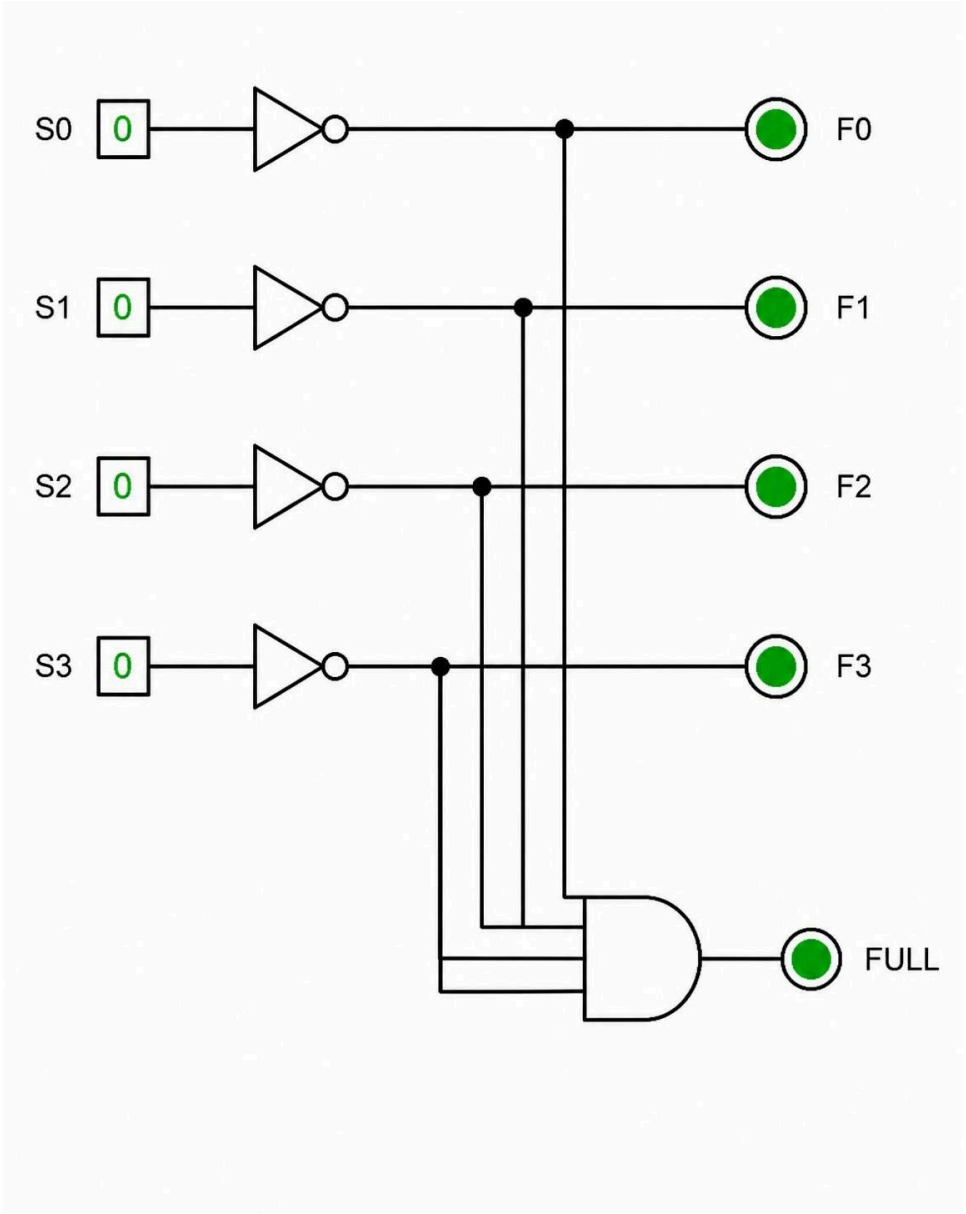
$$F_i = \overline{S_i} \quad \text{for } i \in \{0, 1, 2, 3\}$$

The lot is full when all four bays are simultaneously occupied:

$$\text{FULL} = S_0 \cdot S_1 \cdot S_2 \cdot S_3$$

By De Morgan's laws, any free slot is the inverse of full:

$$\text{ANY_FREE} = \overline{\text{FULL}} = F_0 + F_1 + F_2 + F_3$$



LOGISIM SUB-CIRCUIT: INVERTERS & 4-INPUT AND GAT

Entry control opens the gate only when three conditions hold

Barrier release rule

$$\mathrm{GATE} = V \cdot P \cdot \overline{\mathrm{FULL}}$$

All three predicates must be asserted simultaneously:

$V = 1$: Vehicle present on loop detector
$P = 1$: Valid RFID/permit scanned
$\mathrm{FULL} = 0$: At least one bay available

IMPLEMENTED VIA 3-INPUT AND GATE

V	P	FULL	GATE
1	1	0	**1**

FAIL-SAFE · UNPERMITTED VEHICLES OR FULL LOT LOCK THE BARRIER

A priority encoder points the driver to the nearest free slot

ENCODER STAGE

Priority hierarchy

Slot 0 has highest priority; Slot 3 lowest.

$$Y_1 = \overline{F_0} \cdot \overline{F_1}$$

$$Y_0 = \overline{F_0} \cdot (F_1 + \overline{F_2})$$

K-map minimization shows how the expressions reduce to a few gates.

Free slots	Y1 Y0	Slot shown
F0 free	**00**	Slot 0
F0 busy, F1 free	**01**	Slot 1
F0, F1 busy, F2 free	**10**	Slot 2
Only F3 free	**11**	Slot 3

SCHEMATIC DESIGN

The complete circuit uses basic gate primitives

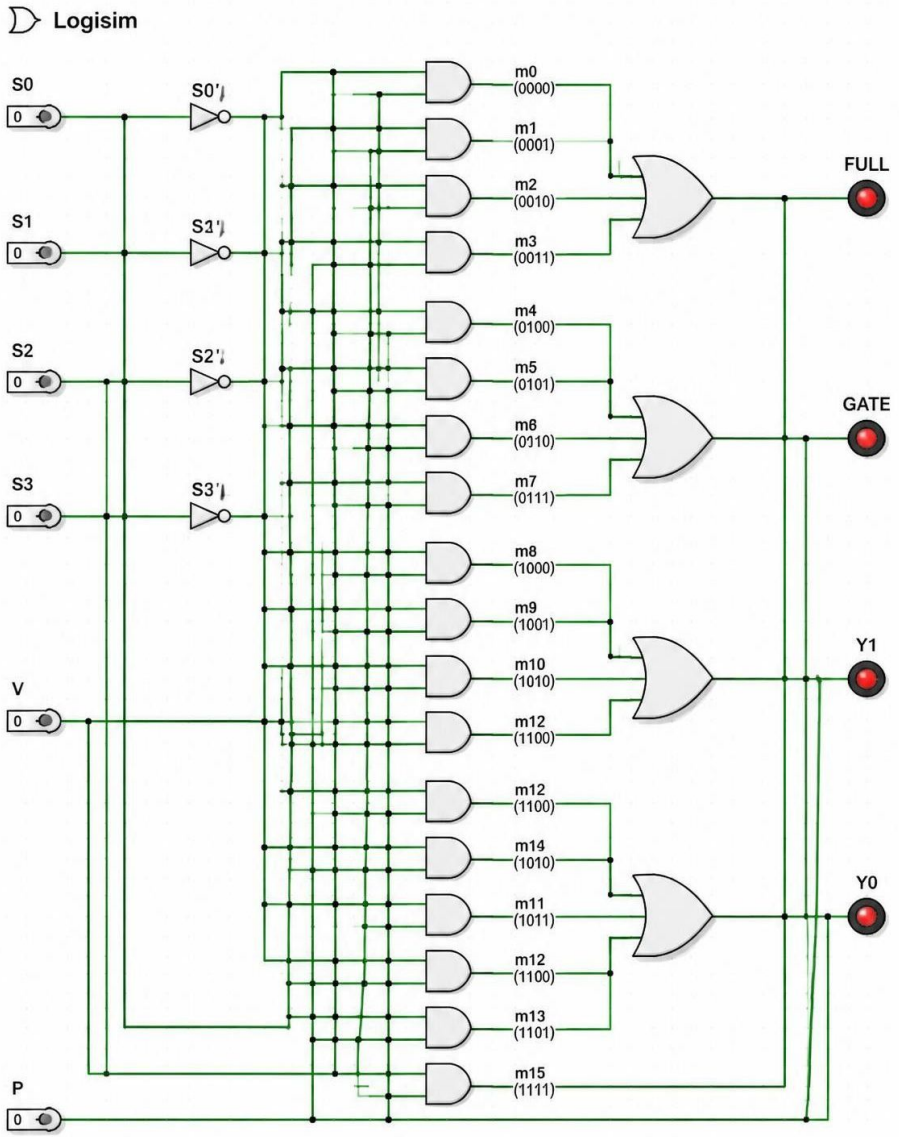
1. **Availability stage** · 4x NOT gates generate active-high slot vacancy flags.

2. **Capacity monitor** · 1x 4-input AND (or NAND/NOT) yields the FULL status flag.

3. **Priority routing** · Two-level AND-OR logic encodes the 2-bit slot coordinate.

4. **Access control** · 1x 3-input AND gate pulses the barrier servo motor output.

TOTAL HARDWARE FOOTPRINT: 3 STANDARD TTL INTEGRATED CIRCUITS



COMPLETE SCHEMATIC WIRING DIAGRAM IN LOGISIM

VERIFICATION

Logisim simulation checks every case that matters

1. **All empty (0000)** · FULL=0; GATE=1 (V=1, P=1); Slot 0 allocated (00).

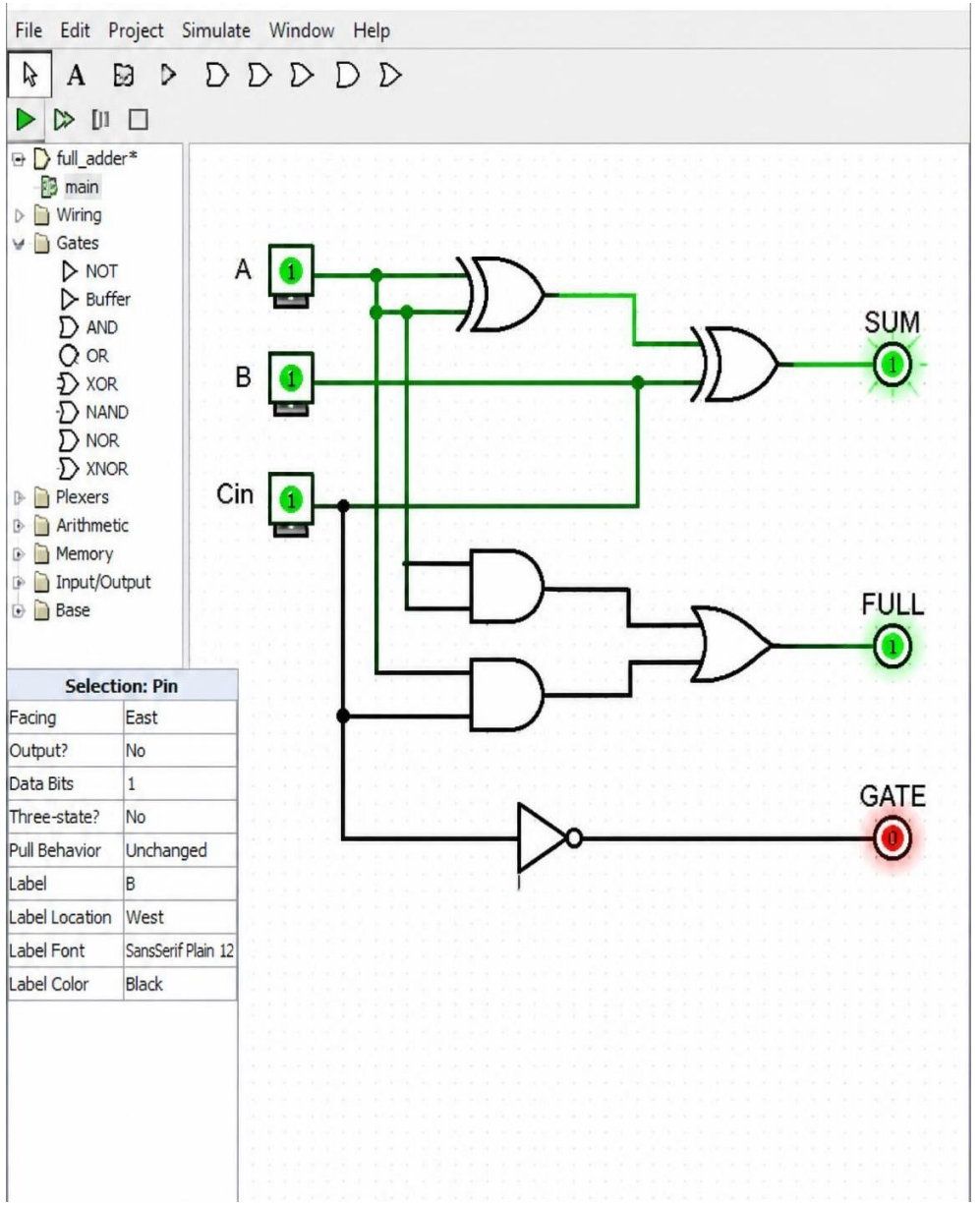
2. **Slot 0 taken (1000)** · Encoder skips to Slot 1 (01); GATE stays enabled.

3. **Only slot 3 free (1110)** · Slot 3 (11); FULL=0; GATE opens.

4. **Lot full (1111)** · FULL=1; GATE=0 regardless of V and P status.

5. **No permit (P=0)** · GATE=0 even with available parking slots.

Compare outputs with the truth table; add results from the student's simulation run.



INTERACTIVE TESTBENCH: INPUT VECTORS VS OUTPUT PROBES

REVIEW & EXTENSIONS

Conclusion and self-check

Design summary

Combinational logic delivers zero-latency slot allocation and secure barrier control with minimal silicon overhead.

Planned extensions

- Scale architecture from 4 slots to N-slot modular cascades
- Add synchronous up/down counter for total occupancy tracking
- Incorporate exit loop sensor and automated exit gate control

Self-check assessment

Q1: Which single gate produces the FULL flag?

Answer: A 4-input AND gate evaluating $S_0 \cdot S_1 \cdot S_2 \cdot S_3$.

Q2: What happens to GATE when permit P = 0?

Answer: GATE is forced low (0) by the 3-input AND gate, barring entry.